

Project 1 : FraudLens360

Combating Fraud in Digital Financial Systems

Insights from the PaySim Dataset



Problem Identification

PROBLEM OVERVIEW

- Digital financial transactions are increasingly vulnerable to fraud.
- Fraudulent behavior can disrupt trust, impact users, and cause monetary losses.
- Traditional detection systems often fail to catch evolving fraud patterns.
- **Goal:** To identify high-risk transaction types, detect suspicious trends and behaviors, and provide data-driven recommendations through visual storytelling using SQL and Power BI – laying the foundation for future fraud detection systems.

Dataset Overview

- Source: PaySim : a synthetic dataset simulating mobile money transactions in a financial ecosystem
- Size: ~1 million transactions (1,048,575 rows)
- Key Columns: type, amount, oldbalanceOrg, newbalanceDest, isFraud, isFlaggedFraud, etc.
- Purpose: Mimics real-world digital financial activity for research in fraud detection.

no fraud was
flagged

statistical
models alone
failed

Losses were
concentrated
but missed

Behavior is
repeated and
templated

Summary

Analysis and Insights

Tools Used: **MySQL** for analysis,
Power BI for visualization

Data
Cleaning



SQL Analysis &
Feature Engg.



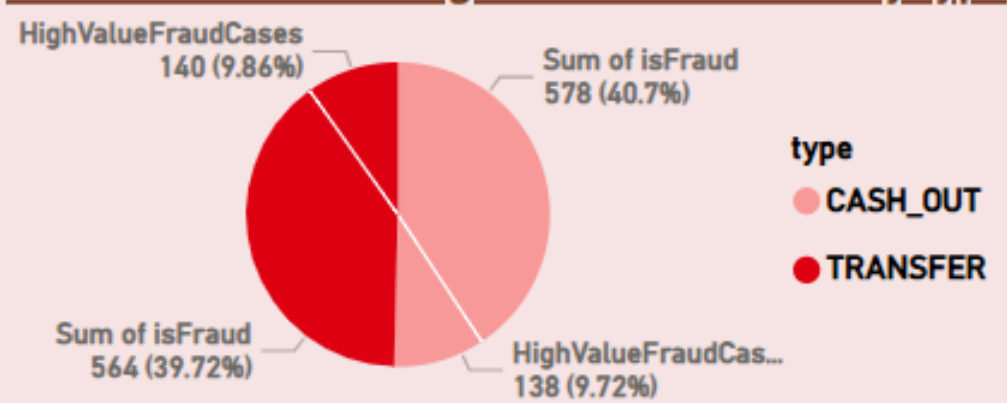
Power BI
Dashboards



Insights & Business
Recommendations

Insights

Sum of isFraud and HighValueFraudCases by type



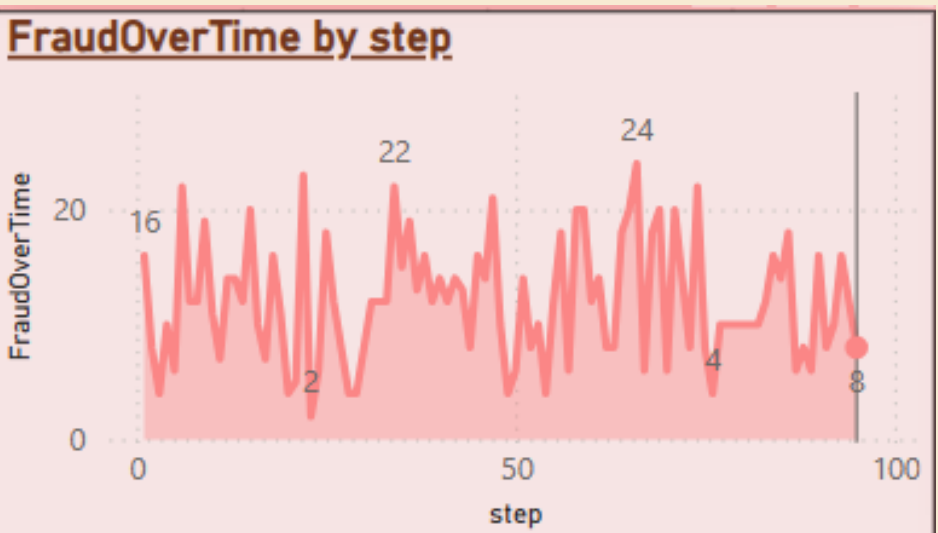
Insight

- Over 99% of fraudulent transactions are found in just two types:
 - CASH_OUT and TRANSFER
- TRANSFER has the highest share of high-value frauds.
- Other transaction types (PAYMENT, DEBIT, CASH_IN) show almost zero fraud.

UPDATE

This gives us a clear starting point. Instead of spreading our efforts thin, we know that if we just focus our fraud detection efforts on these two transaction types, we could catch almost all the fraud.

! Fraud Spike at Step 22 & 66



Insight

- While exploring fraud trends over time, we spotted something strange.
- At two specific time steps – Step 22 and Step 66 – there's a sharp spike in fraudulent activity.
- It's not a gradual rise; it's more like a sudden burst, which clearly stands out from the rest of the timeline.

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UPDATE

COVERING ASPECTS

The System Didn't Catch a Single Fraud

⚠️ "No fraud transactions were flagged—100% detection failure."

Insight: The analysis shows that none of the fraud transactions were flagged by the detection system (`isFlaggedFraud` is always 0). This results in a **0% detection rate**, despite clear instances of fraud (`isFraud` = 1)

📌 This highlights a complete failure of the fraud flagging mechanism—an area that needs critical attention for system improvement.



"Flagging System Failed"

This isn't just a gap, it's a total system failure.

0

Flagged Fraud
Txn

1142

Fraud Txn Count

No alerts, no flags,
no intervention,
even when fraud
was massive.

nameOrig	Total Fraud Amount	Fraud Txn Count
C1079335762	1,00,00,000.00	1
C1237313447	1,00,00,000.00	1
C1438388258	1,00,00,000.00	1
C1439740840	1,00,00,000.00	1
C1843531771	1,00,00,000.00	1
C2050703310	1,00,00,000.00	1
C29118015	1,00,00,000.00	1
C351297720	1,00,00,000.00	1

These Users Account for the **Biggest Fraud Losses**

It challenges the common **assumption that fraud comes from repeat behavior.**

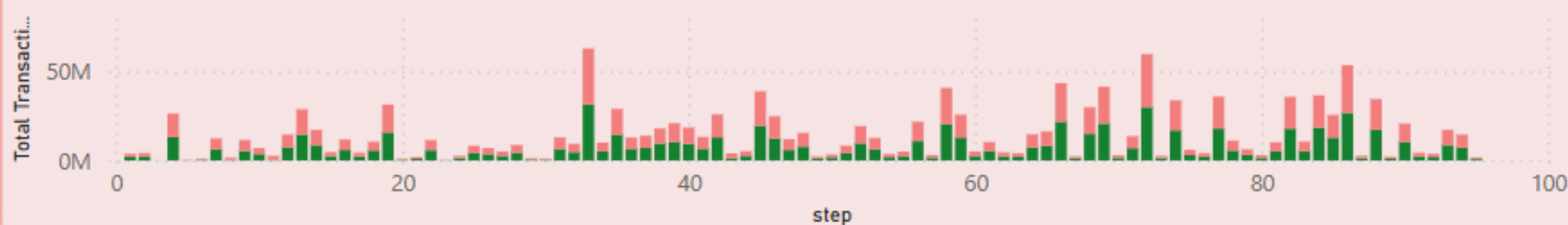
Sometimes, it's a **well-timed, single hit that causes the most loss.**

So systems must flag suspicious behavior even for first-time users, especially when the amount is unusually high.



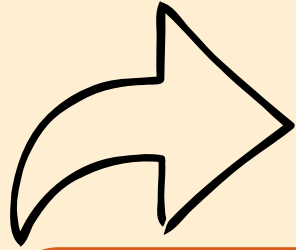
Total Transaction Amount by step and type

type ● CASH_OUT ● TRANSFER



We expected fraud to occur during high-value transaction surges. But interestingly, the biggest spikes in total amount aren't necessarily the steps where most fraud happens.

Fraud hides not just in high-value bursts, but also during normal-looking periods, making it even harder to detect without deeper analysis.



CASHOUT vs TRANSFER



Cashout

680.38M

Total Fraud Loss

1.18M

Avg Fraud Amount

“**Cash Out**” is the final step in many fraudulent chains, once money is cashed out, it's gone.

And yet, not a single one of these transactions was flagged.

This isn't just a miss. It's a systemic blind spot in one of the most critical transaction types.

Transfer

681.60M

Total Fraud Loss

1.21M

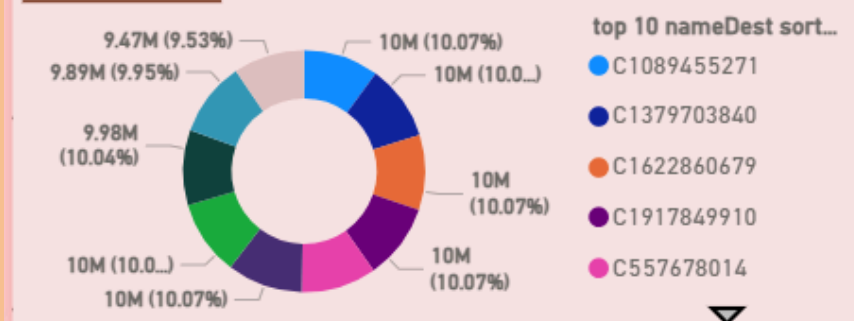
Avg Fraud Amount

“**Transfer**” is the launchpad for most fraudulent flows, the first move in laundering or rerouting money.

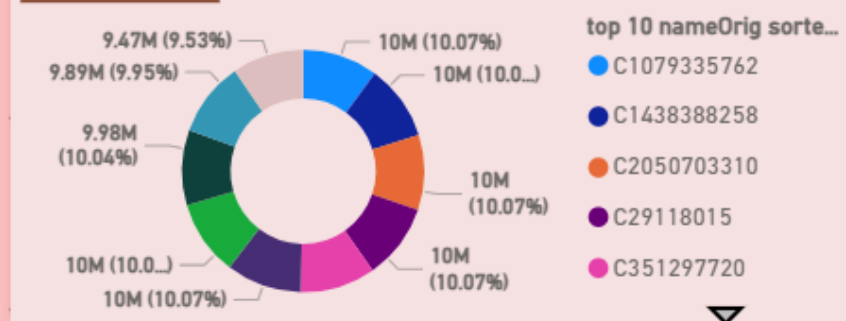
And yet, not a single one of these transactions was flagged.

It's a foundational weakness in the fraud detection pipeline where the attack begins.

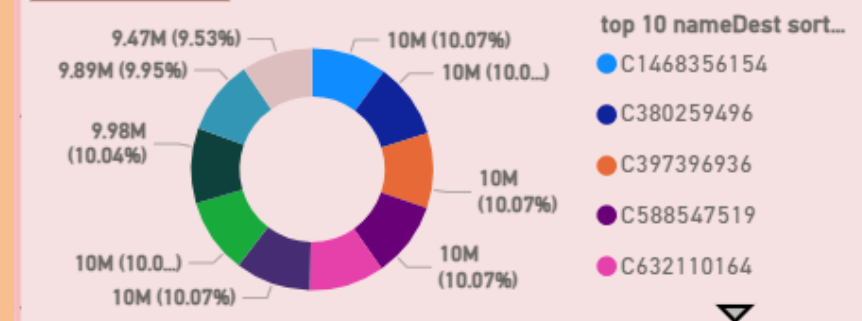
Total Fraud Amount by top 10 nameDest sorted by total fraud amount



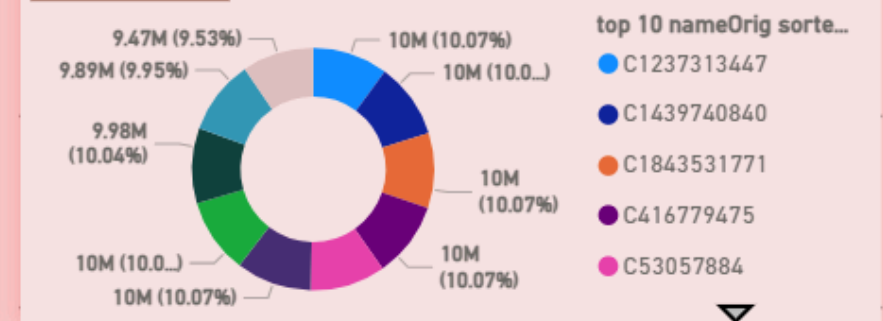
Total Fraud Amount by top 10 nameOrig sorted by total fraud amount



Total Fraud Amount by top 10 nameDest sorted by total fraud amount



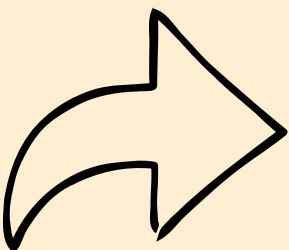
Total Fraud Amount by top 10 nameOrig sorted by total fraud amount



Surprisingly, when we examined who's behind the Cash Out fraud, we didn't find one or two culprits — we found dozens of users, each causing similar levels of loss (~₹10M).

This indicates widespread exploitation of the system — not a few masterminds, but many users abusing it in the same predictable pattern.

When we analyze the top 10 senders (nameOrig) and top 10 receivers (nameDest) in fraudulent Transfer transactions, Each sender and receiver transacts **exactly 10M** or **near-identical values**. Total fraud amount per user is nearly equal, indicating a structured, split-distribution strategy.

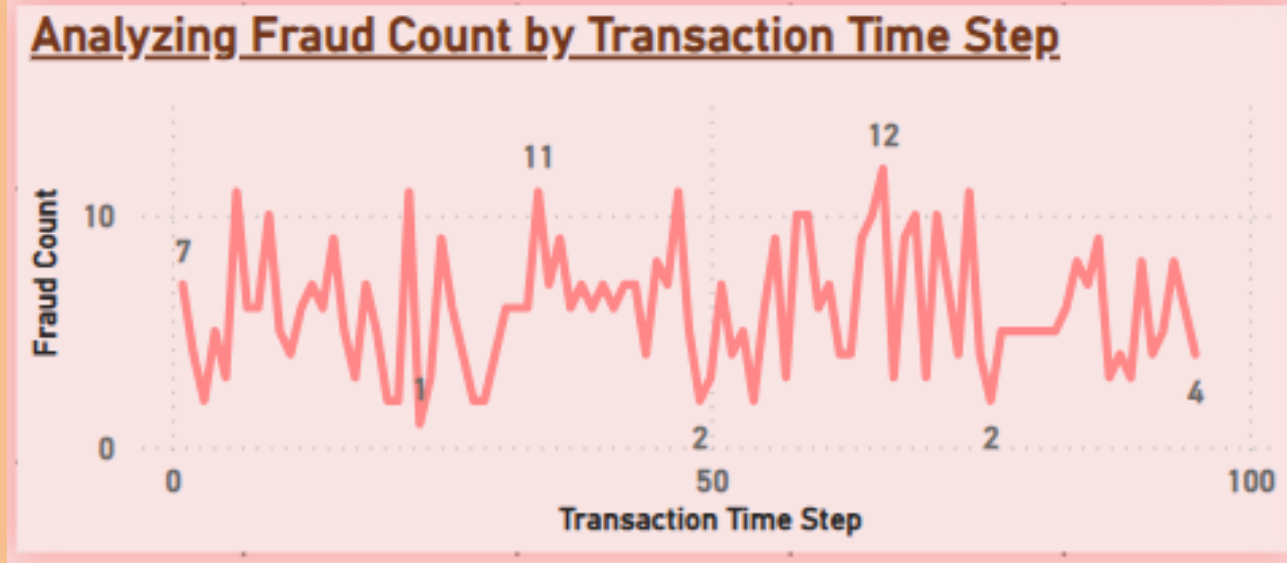
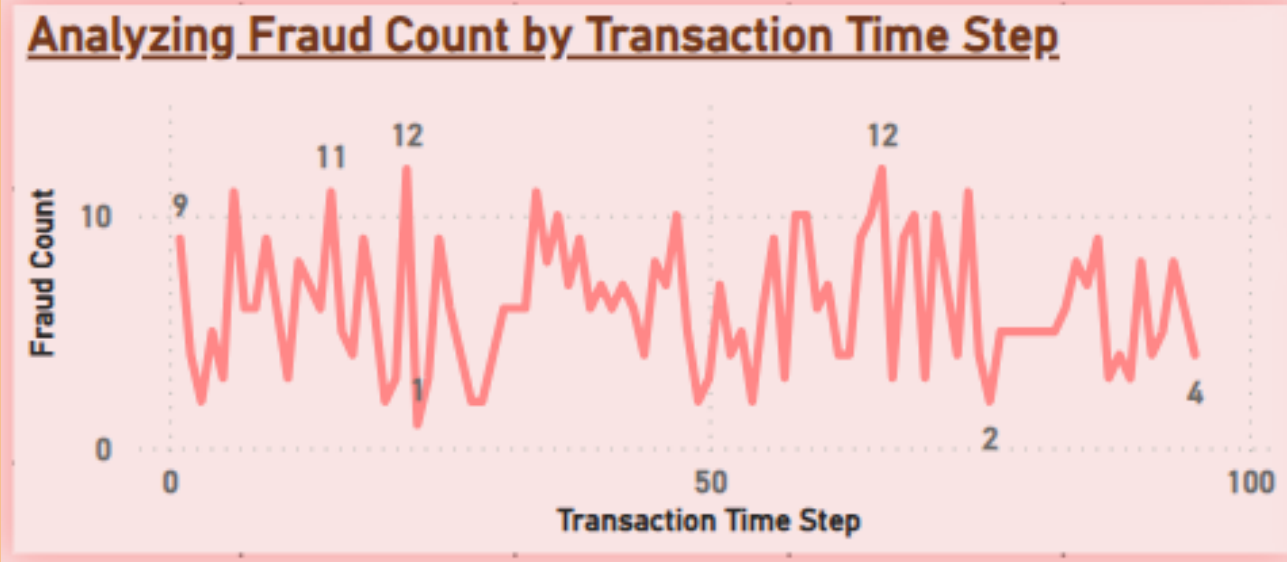


CASHOUT vs TRANSFER



Cashout

Transfer



As fraud analysts, we usually ask “who” or “how.” But in this case, the real power lies in “when.” Our time-series view of Cash Out frauds revealed steep surges in activity at **Steps 6, 15, 22, 34, 47, 58, 59, 66, and 74**, alongside stable fraud bursts at sequences like **7–8, 31–32, 58–59, and 77–82**.

This rhythmic behavior tells us that fraud isn’t chaotic — it’s strategic.

Even when we isolate TRANSFER transactions, we see a familiar pattern — fraud doesn't scatter randomly across time. Instead, it clusters again at **Steps 6, 15, 22, 34, 47, 58–59, and 66** — just like in CashOut. **This isn’t coincidental.** It’s a replication of behavior across transaction types — and likely suggests either: Reuse of the same bot scripts Or a single fraud model operating across multiple channels.

Recommendations

Build a Hybrid Fraud Defense Engine

1. Behavior-Aware Alerting System

Train models not just on fraud counts, but on:

- Timing (steps)
- Repetitive user patterns
- Fixed transaction amounts
- Cross-type replication

This isn’t just anomaly detection, it’s fraud behavior modeling.

2. Auto-Learn from Spikes

Whenever a fraud spike is detected — store its signature (step, amount, user type) and feed it back into alert logic.

-> Create a self-improving fraud firewall.

3. Add Explainable AI Layer

Integrate an Explainable ML model that not only flags a transaction but explains:

- “Why was this flagged?”
- “What’s similar to previous frauds?”
- “Has this pattern been seen in another transaction type?”

THANK YOU

• AYUSH PRAJAPATI

